STRUCTURAL CALCULATION FOR:

735 SF ADU

LAKE ELSINORE, CA

PREPARED BY:

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DECEMBER 01, 2024

FOR USE IN THE CITY OF LAKE ELSINORE

Terence Chan

		WEIGHT (PSF)
ROOF LOAD:		**Asphalt Shingle**
Roofing material		2.5
2X6 Rafter @ 24" O.C.		2.0
1/2" plywood sheathing		2.5
Solar panel		1.0
	DEAD LOAD:	8.0
	LIVE LOAD:	20.0
CEILING LOAD:		
R-30 Insulation		2.5
1/2" Drywall		2.0
Ceiling Joist		2.0
Misc		0.5
	DEAD LOAD:	7.0
	LIVE LOAD:	10.0
FLOOR LOAD:		
2x12 Floor Joist @ 16" O.C.		3.3
5/8" plywood sheathing		1.8
5/8" drywall		2.5
R-19 Insulation.		1.2
Misc		1.2
	DEAD LOAD:	10.0
	LIVE LOAD:	40.0
EXTERIOR WALL:		
2x4 stud @ 16" o.c.		1.0
Stucco		8.0
1/2" drywall		2.5
R-13 Insulation.		2.0
Misc		1.5
	DEAD LOAD:	15.0
INTERIOR WALL:		
2x4 stud @ 16" o.c.		1.0
1/2" drywall (2 layer)		5.0
Misc		1.0
	DEAD LOAD:	7.0

FOR USE IN THE CITY OF LAKE ELSINORE

Location: ADU Rafter
Roof Rafter
Roof Rafter [2021 International Building Code(2018 NDS)
1.5 IN x 5.5 IN x 14.5 FT (12.5 + 2) @ 16 O.C.
#2 - Douglas-Fir-Larch - Dry Use
Section Adequate By: 32.4%
Controlling Factor: Deflection

DEFLECTIONS	Center		Right	
Live Load	0.47	IN L/331	0.03	IN 2L/1802
Dead Load	0.18	in	0.00	in
Total Load	0.65	IN L/238	0.00	IN 2L/∞

Live Load Deflection Criteria: L/240 Total Load Deflection Criteria: L/180

REACTIONS	A		B	
Live Load	167	lb	224	lb
Dead Load	67	lb	92	lb
Total Load	234	lb	316	lb
Bearing Length	0.25	in	0.34	in

SUPPORT LOADS	A		B	
Live Load	125	plf	168	plf
Dead Load	50	plf	69	plf
Total Load	176	plf	237	plf

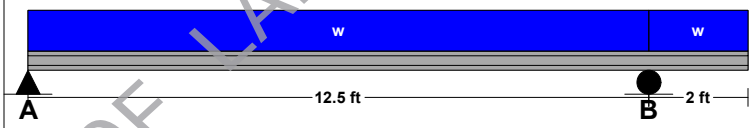
MATERIAL PROPERTIES
#2 - Douglas-Fir-Larch

	Base Values		Adjusted	
Bending Stress:	Fb = 900 psi		Fb' = 1547 psi	
	Cd=1.15 CF=1.30 Cr=1.15			
Shear Stress:	Fv = 180 psi		Fv' = 207 psi	
	Cd=1.15			
Modulus of Elasticity:	E = 1600 ksi		E' = 1600 ksi	
Comp. ⊥ to Grain:	Fc - ⊥ = 625 psi		Fc - ⊥' = 625 psi	

Controlling Moment: 725 ft-lb
6.248 Ft from left support of span 2 (Center Span)
Created by combining all dead loads and live loads on span(s) 2
Controlling Shear: -234 lb
12.612 Ft from left support of span 2 (Center Span)
Created by combining all dead loads and live loads on span(s) 2, 3

Comparisons with required sections:	Req'd	Provided
Section Modulus:	5.62 in3	7.56 in3
Area (Shear):	1.7 in2	8.25 in2
Moment of Inertia (deflection):	15.7 in4	20.8 in4
Moment:	725 ft-lb	975 ft-lb
Shear:	-234 lb	1139 lb

LOADING DIAGRAM



RAFTER DATA	Interior	Eave
Span Length	12.5 ft	2 ft
Rafter Pitch	3 :12	

Roof sheathing applied to top of joists-top of rafters fully braced.

Roof Duration Factor	1.15
Peak Notch Depth	0.00
Base Notch Depth	0.00

RAFTER LOADING

Uniform Roof Loading

Roof Live Load:	LL =	20	psf
Roof Dead Load:	DL =	8	psf
Non-Snow Roof Loaded Area:	RLA =	0	psf

Slope Adjusted Spans And Loads

Interior Span:	L-adj =	12.88	ft
Eave Span:	L-Eave-adj =	2.06	ft
Interior Live Load:	wL-adj =	25	plf
Eave Live Load:	wL-Eave-adj =	25	plf
Interior Dead Load:	wD-adj =	10	plf
Eave Dead Load:	wD-Eave-adj =	10	plf
Interior Total Load:	wT-adj =	35	plf
Eave Total Load:	wT-Eave-adj =	35	plf

FOR USE IN THE CITY OF LAKE ELSINORE

Location: Main House Ceil Joist
Floor Joist
Floor Joist [2021 International Building Code(2018 NDS)
1.5 IN x 5.5 IN x 11.67 FT @ 16 O.C.
#2 - Douglas-Fir-Larch - Dry Use
Section Adequate By: 36.9%
Controlling Factor: Deflection

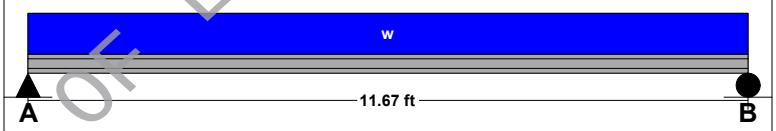
DEFLECTIONS	Center	
Live Load	0.17	IN L/838
Dead Load	0.12	in
Total Load	0.28	IN L/493

Live Load Deflection Criteria: L/240 Total Load Deflection Criteria: L/360

REACTIONS	A		B	
Live Load	78	lb	78	lb
Dead Load	54	lb	54	lb
Total Load	132	lb	132	lb
Bearing Length	0.14	in	0.14	in

SUPPORT LOADS	A		B	
Live Load	59	plf	59	plf
Dead Load	41	plf	41	plf
Total Load	99	plf	99	plf

LOADING DIAGRAM



MATERIAL PROPERTIES
#2 - Douglas-Fir-Larch

	Base Values			Adjusted		
Bending Stress:	Fb =	900	psi	Fb' =	1346	psi
	Cd=1.00 CF=1.30 Cr=1.15					
Shear Stress:	Fv =	180	psi	Fv' =	180	psi
	Cd=1.00					
Modulus of Elasticity:	E =	1600	ksi	E' =	1600	ksi
Comp. ⊥ to Grain:	Fc - ⊥ =	625	psi	Fc - ⊥' =	625	psi

Controlling Moment: 386 ft-lb
5.84 Ft from left support of span 2 (Center Span)
Created by combining all dead loads and live loads on span(s) 2
Controlling Shear: -132 lb
12.0 Ft from left support of span 2 (Center Span)
Created by combining all dead loads and live loads on span(s) 2

Comparisons with required sections:	Req'd		Provided	
Section Modulus:	3.44	in3	7.56	in3
Area (Shear):	1.1	in2	8.25	in2
Moment of Inertia (deflection):	15.2	in4	20.8	in4
Moment:	386	ft-lb	848	ft-lb
Shear:	-132	lb	990	lb

JOIST DATA	Center	
Span Length	11.67	ft
Unbraced Length-Top	0	ft
Unbraced Length-Bottom	0	ft

Floor sheathing applied to top of joists-top of joists fully braced.
Floor Duration Factor 1.00

JOIST LOADING

Uniform Floor Loading		Center	
Live Load	LL =	10	psf
Dead Load	DL =	7	psf
Total Load	TL =	17	psf
TL Adj. For Joist Spacing	wT =	22.7	plf

FOR USE IN THE CITY OF LAKE ELSINORE

Location: CB-1
Uniformly Loaded Floor Beam
Uniformly Loaded Floor Beam [2021 International Building Code(2018 NDS)
3.5 IN x 9.25 IN x 13.2 FT
#1 - Douglas-Fir-Larch - Dry Use
Section Adequate By: 0.5%
Controlling Factor: Moment

DEFLECTIONS Center

Live Load	0.23	IN L/700
Dead Load	0.17	in
Total Load	0.40	IN L/399

Live Load Deflection Criteria: L/240 Total Load Deflection Criteria: L/240

REACTIONS

	A		B	
Live Load	858	lb	858	lb
Dead Load	647	lb	647	lb
Total Load	1505	lb	1505	lb
Bearing Length	0.69	in	0.69	in

BEAM DATA Center

Span Length	13.2	ft
Unbraced Length-Top	0	ft
Floor Duration Factor	1.00	
Notch Depth	0.00	

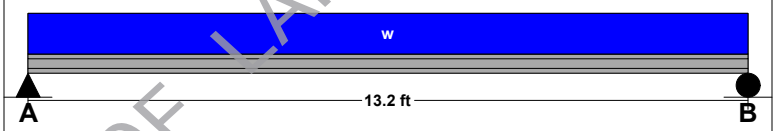
MATERIAL PROPERTIES
#1 - Douglas-Fir-Larch

	Base Values		Adjusted	
Bending Stress:	Fb = 1000	psi	Fb' = 1200	psi
	Cd=1.00 CF=1.20			
Shear Stress:	Fv = 180	psi	Fv' = 180	psi
	Cd=1.00			
Modulus of Elasticity:	E = 1700	ksi	E' = 1700	ksi
Comp. ⊥ to Grain:	Fc - ⊥ = 625	psi	Fc - ⊥' = 625	psi

Controlling Moment: 4966 ft-lb
6.6 ft from left support
Created by combining all dead and live loads.
Controlling Shear: -1505 lb
At support.
Created by combining all dead and live loads.

Comparisons with required sections:	Req'd		Provided	
Section Modulus:	49.66	in3	49.91	in3
Area (Shear):	12.54	in2	32.38	in2
Moment of Inertia (deflection):	138.8	in4	230.84	in4
Moment:	4966	ft-lb	4991	ft-lb
Shear:	-1505	lb	3885	lb

LOADING DIAGRAM



FLOOR LOADING

			Side 1		Side 2	
Floor Live Load	FLL =	10	psf	0	psf	
Floor Dead Load	FDL =	7	psf	0	psf	
Floor Tributary Width	FTW =	13	ft	0	ft	
Wall Load	WALL =	0	plf			

BEAM LOADING

Beam Total Live Load:	wL =	130	plf
Beam Total Dead Load:	wD =	91	plf
Beam Self Weight:	BSW =	7	plf
Total Maximum Load:	wT =	228	plf

FOR USE IN THE CITY OF LAKE ELSINORE

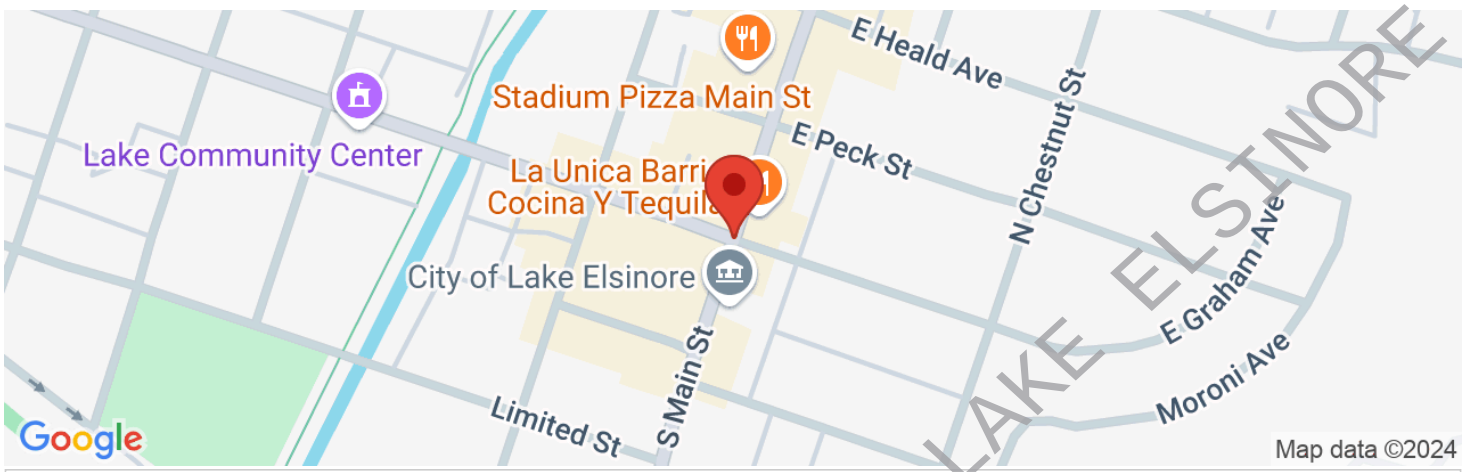
USGS web services were down for some period of time and as a result this tool wasn't operational, resulting in *timeout* error.
USGS web services are now operational so this tool should work as expected.

OSHPD

ADU

Lake Elsinore, CA, USA

Latitude, Longitude: 33.6680447, -117.3276917



Date	12/3/2024, 5:23:53 PM
Design Code Reference Document	ASCE7-16
Risk Category	II
Site Class	D - Default (See Section 11.4.3)

Type	Value	Description
S_S	1.969	MCE_R ground motion. (for 0.2 second period)
S_1	0.712	MCE_R ground motion. (for 1.0s period)
S_{MS}	2.363	Site-modified spectral acceleration value
S_{M1}	null -See Section 11.4.8	Site-modified spectral acceleration value
S_{DS}	1.575	Numeric seismic design value at 0.2 second SA
S_{D1}	null -See Section 11.4.8	Numeric seismic design value at 1.0 second SA

Type	Value	Description
SDC	null -See Section 11.4.8	Seismic design category
F_a	1.2	Site amplification factor at 0.2 second
F_v	null -See Section 11.4.8	Site amplification factor at 1.0 second
PGA	0.844	MCE_G peak ground acceleration
F_{PGA}	1.2	Site amplification factor at PGA
PGA_M	1.013	Site modified peak ground acceleration
T_L	8	Long-period transition period in seconds
SsRT	1.969	Probabilistic risk-targeted ground motion. (0.2 second)
SsUH	2.18	Factored uniform-hazard (2% probability of exceedance in 50 years) spectral acceleration
SsD	2.352	Factored deterministic acceleration value. (0.2 second)
S1RT	0.712	Probabilistic risk-targeted ground motion. (1.0 second)
S1UH	0.79	Factored uniform-hazard (2% probability of exceedance in 50 years) spectral acceleration.
S1D	0.938	Factored deterministic acceleration value. (1.0 second)
PGAd	0.991	Factored deterministic acceleration value. (Peak Ground Acceleration)
PGA_{UH}	0.844	Uniform-hazard (2% probability of exceedance in 50 years) Peak Ground Acceleration
C_{RS}	0.903	Mapped value of the risk coefficient at short periods
C_{R1}	0.901	Mapped value of the risk coefficient at a period of 1 s
C_V	1.494	Vertical coefficient

FOR USE IN THE CITY OF LAKE ELSINORE

FOR USE IN THE CITY OF LAKE ELSINORE

Table 27.6-1
MWFRS – Part 2: Wind Loads – Walls
Exposure B

V(mph)	110			115			120			130			140			160			180			200		
h(ft.), L/B	0.5	1	2	0.5	1	2	0.5	1	2	0.5	1	2	0.5	1	2	0.5	1	2	0.5	1	2	0.5	1	2
160	38.1	37.7	34.1	42.1	41.7	37.8	46.4	45.9	41.7	55.8	55.1	50.2	66.3	65.4	59.7	91.0	89.4	81.8	120.8	118.3	108.5	156.2	152.4	140.0
	25.6	25.4	21.0	28.3	28.1	23.3	31.2	30.9	25.7	37.5	37.1	30.9	44.6	44.0	36.8	61.2	60.1	50.4	81.3	79.6	66.9	105.2	102.6	86.2
150	36.9	36.6	33.0	40.7	40.4	36.5	44.9	44.4	40.3	53.9	53.3	48.5	63.9	63.1	57.6	87.5	86.1	78.9	116.1	113.8	104.5	149.9	146.5	134.7
	25.1	24.9	20.6	27.7	27.5	22.8	30.5	30.2	25.2	36.7	36.2	30.3	43.5	43.0	36.0	59.6	58.6	49.3	79.0	77.4	65.3	102.0	99.7	84.2
140	35.6	35.4	31.9	39.3	39.1	35.3	43.3	42.9	38.9	51.9	51.4	46.7	61.5	60.8	55.5	84.0	82.8	75.9	111.2	109.2	100.4	143.5	140.5	129.3
	24.5	24.4	20.2	27.1	26.9	22.4	29.8	29.6	24.6	35.7	35.4	29.6	42.4	41.9	35.2	57.9	57.0	48.1	76.6	75.2	63.7	98.8	96.7	82.0
130	34.4	34.2	30.8	37.9	37.7	34.0	41.7	41.4	37.4	49.9	49.5	44.9	59.1	58.5	53.3	80.5	79.5	72.8	106.3	104.6	96.2	136.9	134.3	123.8
	24.0	23.9	19.8	26.5	26.3	21.9	29.1	28.9	24.1	34.8	34.5	28.9	41.2	40.8	34.3	56.2	55.4	46.9	74.2	73.0	62.0	95.5	93.7	79.8
120	33.1	33.0	29.6	36.5	36.3	32.7	40.1	39.9	35.9	47.9	47.6	43.1	56.6	56.2	51.0	76.9	76.1	69.6	101.3	99.9	91.8	130.2	128.0	118.0
	23.4	23.3	19.4	25.8	25.7	21.4	28.4	28.2	23.6	33.9	33.7	28.3	40.1	39.7	33.5	54.4	53.8	45.6	71.7	70.7	60.2	92.2	90.6	77.4
110	31.8	31.7	28.4	35.1	34.9	31.3	38.5	38.3	34.4	45.9	45.6	41.2	54.1	53.8	48.8	73.3	72.6	66.3	96.3	95.1	87.4	123.5	121.6	112.1
	22.9	22.8	19.0	25.2	25.1	20.9	27.7	27.5	23.0	33.0	32.8	27.6	38.9	38.7	32.6	52.7	52.2	44.4	69.2	68.4	58.4	88.8	87.4	75.0
100	30.5	30.4	27.1	33.6	33.5	29.9	36.8	36.7	32.9	43.8	43.6	39.3	51.6	51.3	46.4	69.6	69.1	62.9	91.2	90.3	82.8	116.6	115.1	106.0
	22.3	22.3	18.5	24.6	24.5	20.4	26.9	26.8	22.5	32.1	31.9	26.8	37.8	37.6	31.7	50.9	50.5	43.0	66.7	66.0	56.6	85.3	84.2	72.5
90	29.2	29.1	25.9	32.1	32.0	28.5	35.1	35.0	31.2	44.7	41.6	37.3	49.1	48.8	44.0	65.9	65.5	59.5	86.0	85.3	78.0	109.6	108.5	99.8
	21.8	21.7	18.1	23.9	23.9	19.9	26.2	26.1	21.9	31.1	31.0	26.1	36.6	36.4	30.8	49.2	48.9	41.7	64.2	63.6	54.6	81.8	80.9	69.9
80	27.8	27.7	24.5	30.5	30.5	27.0	33.4	33.3	29.6	39.6	39.5	35.2	46.4	46.3	41.5	62.2	61.9	55.9	80.8	80.3	73.1	102.6	101.7	93.3
	21.2	21.2	17.7	23.3	23.2	19.4	25.5	25.4	21.3	30.2	30.1	25.4	35.4	35.3	29.9	47.4	47.2	40.3	61.6	61.2	52.6	78.3	77.6	67.2
70	26.3	26.3	23.1	28.9	28.8	25.4	31.6	31.5	27.9	37.4	37.3	33.1	43.7	43.6	38.9	58.3	58.1	52.2	75.5	75.1	68.1	95.5	94.9	86.6
	20.6	20.6	17.2	22.6	22.6	18.9	24.7	24.7	20.7	29.3	29.2	24.6	34.2	34.2	28.9	45.6	45.5	38.8	59.1	58.8	50.6	74.7	74.3	64.3
60	24.8	24.8	21.7	27.2	27.1	23.8	29.7	29.6	26.1	35.1	35.0	30.9	41.0	40.9	36.2	54.4	54.2	48.4	70.1	69.8	62.8	88.2	87.9	79.6
	20.0	20.0	16.7	21.9	21.9	18.4	23.9	23.9	20.1	28.3	28.2	23.6	33.0	33.0	27.9	43.9	43.8	37.3	56.5	56.3	48.5	71.2	70.9	61.4
50	23.1	23.1	20.2	25.3	25.3	22.1	27.6	27.6	24.2	32.6	32.6	28.6	38.0	38.0	33.4	50.3	50.2	44.5	64.5	64.4	57.4	80.9	80.7	72.5
	19.3	19.3	16.3	21.2	21.2	17.8	23.1	23.1	19.5	27.3	27.3	23.0	31.8	31.8	26.9	42.0	42.0	35.8	54.0	53.8	46.3	67.6	67.5	58.4
40	21.5	21.5	18.6	23.5	23.5	20.4	25.6	25.6	22.3	30.2	30.2	26.3	35.1	35.1	30.7	46.3	46.2	40.7	59.2	59.1	52.3	73.9	73.8	65.7
	18.8	18.7	15.8	20.5	20.5	17.4	22.4	22.4	18.9	26.4	26.4	22.4	30.7	30.7	26.1	40.5	40.4	34.6	51.7	51.7	44.5	64.6	64.5	55.8
30	19.6	19.6	16.9	21.4	21.4	18.5	23.3	23.3	20.2	27.5	27.4	23.8	31.9	31.9	27.7	41.9	41.9	36.6	53.4	53.4	46.8	66.5	66.4	58.5
	18.1	18.1	15.4	19.8	19.8	16.8	21.5	21.5	18.4	25.3	25.3	21.6	29.5	29.5	25.2	38.7	38.7	33.2	49.3	49.3	42.5	61.4	61.3	53.1
20	17.5	17.5	15.1	19.2	19.2	16.6	20.9	20.9	18.1	24.5	24.5	21.2	28.5	28.5	24.7	37.3	37.3	32.4	47.4	47.4	41.3	58.8	58.8	51.4
	17.2	17.2	14.8	18.8	18.8	16.2	20.5	20.5	17.7	24.1	24.1	20.8	28.0	28.0	24.2	36.7	36.7	31.7	46.6	46.6	40.4	57.8	57.7	50.3
15	16.7	16.7	14.5	18.2	18.2	15.8	19.9	19.9	17.3	23.3	23.3	20.3	27.1	27.1	23.6	35.4	35.4	30.9	44.9	44.9	39.3	55.6	55.6	48.7
	16.7	16.7	14.5	18.2	18.2	15.8	19.9	19.9	17.3	23.3	23.3	20.3	27.1	27.1	23.6	35.4	35.4	30.9	44.9	44.9	39.3	55.6	55.6	48.7

SIMPLIFIED METHOD FOR LOW RISE BUILDINGS WITH PROVISIONS OF ASCE 7-10

SEISMIC DESIGN:-

$W = [(8+9)785 + 2(4)(15+10)(35.5)]1.3$
$= 26,580\ \#$

$V = \frac{S_{DS}}{(\frac{R}{I})} W \times 0.7$
$= \frac{1.575}{6.5} \times 0.7\,(26,580)$
$= 4508\ \#$

WIND ANALYSIS:-

$V = 16.7(4+4.5)(35.5)$
$= 5040\ \#$ WIND GOVERNS #

SHEAR WALL DESIGN:-

LINE ① $= 5040/2(9.5)$
$= 266\ \#/'$ — ①

LINE ② $= 5040/2(5+7)$
$= 210\ \#/'$ — ①

LINE ③ $= 5040/2(10)$
$= 252\ \#/'$ — ①

LINE ④ $= 5040/35.5 \times \frac{17}{2}/10$
$= 120\ \#/'$ — ①

LINE ⑤ $= \frac{5040}{35.5} \times \frac{18}{2}/10$
$= 120\ \#/'$ — ①

UPLIFT $= 260 \times 1.25 \times 8$
$= 2600\ \#$
USE HDU2

FOR USE IN THE CITY OF LAKE ELSINORE